

# TrES-5b

R-1422

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_131007 · created 2026-08-01T06:23:32+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2456572.7266 +/- 0.0015 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.039 +/- 0.036                 |
| Transit depth (Rp/Rs)^2                           | 0.15 +/- 0.28 [%]               |
| Orbital Inclination (inc)                         | 88.79 +/- 3.15                  |
| Airmass coefficient 1 (a1)                        | 1.118 +/- 0.024                 |
| Airmass coefficient 2 (a2)                        | -0.083 +/- 0.018                |
| Scatter in the residuals of the lightcurve fit is | 2.18 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.98                            |
| Optimal Annulus                                   | 10.54                           |
| Transit Duration (day)                            | 0.0745 +/- 0.0089               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.039 ± 0.036 vs 0.142 (deviation 2.86σ) |
| Duration fit vs published | 0.0745 d vs 0.0758 d (deviation 0.15σ)   |
| Mid-time O–C              | 2.0 min                                  |
| Depth SNR                 | 1.1                                      |
| Residual scatter          | 2.18 %                                   |

### Light curve

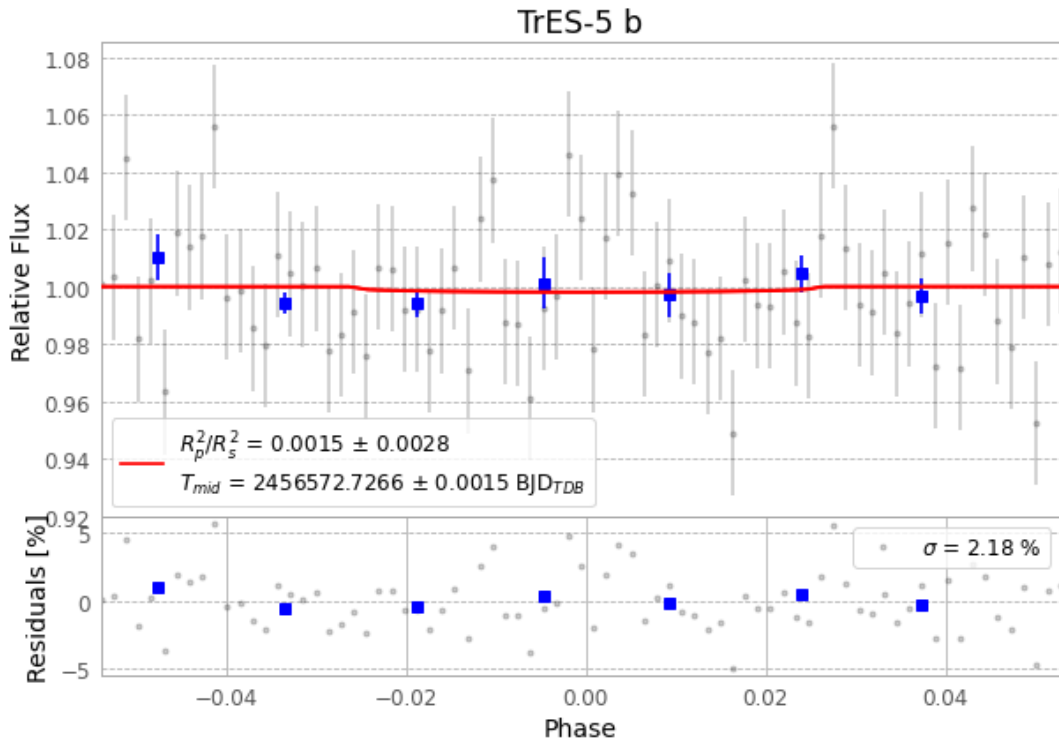


Figure 1 — FinalLightCurve\_TrES-5 b\_2013-10-06.png (SHA-256 e758fe0caa5ff7bd...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 302], 2 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0ffd05106f1b3eae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

77 FITS frames (51 MB), TRES-5131007032712.FITS → TRES-5131007071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-131007.log (e4cb11c89a44...), FinalLightCurve\_TrES-5 b\_2013-10-06.png (e758fe0caa5f...), FinalLightCurve\_TrES-5 b\_2013-10-06.csv (a9ff166800dd...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 06:23Z.